LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	38
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and principle of operation of yaw damping circuit, explanation of torque limiting and phase advance.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and principle of operation of yaw damping circuit, explanation of torque limiting and phase advance.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagrams and signal flow charts; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the yaw damping system?
(b) How does torque reaction affect aircraft heading during power changes?
(c) Why is phase advance necessary in yaw damping circuits?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of yaw damping circuit • Principle of operation • Torque limiting mechanism • Phase advance concept	L & Ds	PPT, IB	15 min
(b)	• Circuit component interaction • Signal processing flow • Torque limit calculation • Phase shift compensation	L & Ds	PPT, PDF	20 min
(c)	• System response characteristics • Failure mode effects • Integration with AFCS • Operational limitations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how torque limiting prevents excessive rudder movement.
(b) Describe the role of phase advance in improving yaw damping response.
(c) Identify the key components involved in the yaw damping circuit operation.

12. Home Assignment and Follow Up	Prepare a brief note on how the yaw damping system interfaces with the artificial feel system discussed in LP 39.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of artificial feel system and helicopter trim system.	Refer to the master syllabus reference list	LP 39

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min